BTS –IV –04.16 – 0446 Reg. No. | | | | | | | | |

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B. Tech. Degree IV Semester Examination April 2016

CS 1403 COMPUTER ARCHITECTURE AND ORGANISATION

(*2012 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Write a short note on instruction sequencing.
(b) Describe Assembler Directives with example.
(c) Explain Booth algorithm with the help of an example.
(d) Describe various steps in executing a complete instruction.
(e) Explain static memory and dynamic memory.
(f) What do you mean by memory interleaving? Explain.
(g) Describe various bus arbitration mechanisms.
(h) Write a short note on interrupt nesting.

PART B

(4 × 15 = 60)

II. (a) Explain the basic operational concepts for executing a typical instruction, ADD LOC A,R0. (7)
(b) Describe various operations to be performed while calling a subroutine. (8)

OR

III. Define addressing mode. Describe various addressing modes in modern processors with examples. (15)

IV. Explain the design and operation of a microprogrammed control unit in detail. (15)

OR

V. (a) Explain the operation of a 4 bit carry-lookahead adder. (7)
(b) Explain the fast multiplication algorithm of 'Bit' pair recoding. (8)

VI. Explain cache memory and various cache mapping functions in detail. (15)

OR

VII. (a) Explain in detail virtual memory and address translation. (10)
(b) Write a short note on secondary storage devices. (5)

VIII. (a) Explain the detailed operation of DMA. (10)
(b) Write a note on vectored interrupts. (5)

OR

IX. Write short notes on: (15)

(i) Synchronous bus.
(ii) Asynchronous bus.
(iii) USB.

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